HU Shicheng

Email: shu40@illinois.edu

EDUCATION

University of Illinois Urbana-Champaign, Urbana-Champaign, IL, USA **08/2026 – Present**
Degree: Master of Computer Science (MCS)
Tongji University, Shanghai, China **09/2022 – 07/2026**
Degree: BE in Computer Science and Technology

RESEARCH INTERESTS

Agentic AI · Trustworthy & Controllable LLMs · Multi-Agent Systems · LLM-based Simulation

• **Persistent Autonomous Agents:** How to transform LLMs into persistent agents with identity, memory, internal state, goals, and the ability to act within environments and interact with other agents.
• **Human Behavior Simulation:** When and to what extent an LLM-based agent can faithfully model human behavior, including whether it can consistently represent a specific individual.
• **Emergent Multi-Agent Behavior:** How interactions among many individually simple agents can produce collective, emergent, system-level behavior.
• **Long-term Goal:** Develop reliable LLM-based simulation frameworks for modeling complex human and social systems.

RESEARCH EXPERIENCES

White-box auditing of large language model unlearning **05/2025 – 08/2026**
Research Assistant, Supervisor: Prof. Mao Yongyi, University of Ottawa
• Proposed a white-box auditing framework, constructed a synthetic Fake Personal Information dataset, and designed the RIG decoding strategy to recover supposedly erased information by exploring low-probability logits.
• Experimented on DeepSeek-7B with 5 unlearning algorithms; found 97% recoverable forgotten content, revealing current methods only suppress surface access to sensitive data.
• Led experimental coding and evaluation pipeline; co-authored the resulting paper, available on arXiv.

Computational Property of Organized Complexity Simulation **02/2025 – 06/2026**
Independent Research, Supervisor: Prof. Wang Cheng, Tongji University
• Developed a theoretical framework linking computational properties with simulation methods, supporting organized complexity modeling.
• Derived mathematical models and proofs to define computational constraints, integrating joint embeddings and machine learning for implementable model structures.
• Designed experiments, implemented algorithms, and authored the full manuscript; Wrote it up as my undergraduate graduation project.

Future Scientific Paradigms in Aerospace Systems (Published) **07/2024 – 12/2024**
Research Assistant, Supervisor: Prof. Wang Cheng, Tongji University
• Researched scientific paradigm evolution and its application in aerospace, focusing on smart fibers, data constraints, and computational challenges;
• Led perspective writing, outlining the future paradigm that unifies materials, aerospace, and information for risk prediction and deep space exploration;
• Co-authored a perspective article on interdisciplinary paradigms, published in National Science Report (3rd author).

TECHNICAL SKILLS

Programming Languages: Python, C++
AI/ML: Multi-Agent Systems, LLM Inference & Evaluation, Model Unlearning
Tools: Git, Linux, Protocol Buffers, OpenGL

INTERNSHIP

NIO Inc., Shanghai, China **07/2025 – 11/2025**

System AI Engineer Intern

• Conducted data preprocessing for path-planning models, addressing model limitations and improving alignment with real-world driving scenarios by adjusting input features to induce safer and more realistic routing strategies.

• Independently developed automated testing tools to evaluate information communication and robustness across complex autonomous driving model chains, significantly improving testing efficiency compared to manual sample code approaches.

• Researched recent reinforcement learning (RL) applications in autonomous driving and trained path-planning models to better reflect natural driver behavior patterns.

COURSE PROJECTS

"InnBook" – LLM-AI-Agent Powered Creative Interaction Platform **05/2025 – 12/2025**

• Developed an immersive creation and interaction platform driven by AI agents, integrating creative writing, tabletop role-playing, simulation, and community engagement into a unified content co-creation ecosystem.

• Built an initial working prototype and iteratively improved AI-agent orchestration, focusing on user experience, narrative coherence, and scalability of multi-agent collaboration.

• Designed interactive storytelling mechanisms and real-time simulation modules, enabling users to generate and evolve storylines collaboratively.

Offline Ray-Traced Fluid Simulation (C++/OpenGL) **10/2024 – 12/2024**

Computer Graphics

• APIC Fluid Dynamics: Implemented APIC fluid dynamics with multi-threaded parallelism, integrating scene meshes through voxelization and BFS-based external marking;

• Fluid-Rigid Interaction: Integrated Bullet Physics for buoyancy dynamics, enabling fluid-rigid body interaction;

• Rendering Optimization: Separated simulation, rendering, and physics processes via thread decoupling and synchronization, achieving a 40% reduction in offline rendering time.

AWARDS

2024 H Prize in the American College Student Mathematical Modeling Competition (Team Leader)

2023 and 2024 Second Prize in the National College Student Mathematical Modeling Competition, Shanghai Division (Two times, Team Leader)

2023 First Prize in the National College Student Mathematics Competition, Shanghai Division

2023-2024 Outstanding Student of Tongji University

2023-2024 Tongji University Scholarship

2022-2023 National Scholarship (Top 2% GPA Awarded)

EXTRACURRICULAR ENGAGEMENT

Illini Rowing | Rower **09/2026 – Present**

Tongji University Dragon Boat Team **02/2024 –06/2026**